Are Headtracking Measures an Indicator of Depressive Symptoms?

360 Videos VR Experiment

Team: Data Analysis Services

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GitHub: github.com/bharath-gajawada/Data_Analysis_Services

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1. Introduction

1.1 Background and Motivation

Depression is one of the most common mental health conditions worldwide, but screening still relies mostly on self-reported questionnaires like the PHQ-9. While these tools are well-validated, they are subjective. They depend on how well people remember and report their own feelings, and they cannot capture real-time behavioral responses (Kroenke et al., 2001). This has created growing interest in finding objective behavioral markers that could support or supplement traditional screening.

One promising idea is to look at psychomotor retardation, which is a core feature of depression involving slowed movements and reduced engagement. Virtual Reality (VR) provides a natural way to measure this. In a 360 degree video environment, people explore their surroundings by moving their heads, and we can precisely track how actively they do this using headtracking data.

1.2 Reference Study

This study is based on the work of Srivastava and Lahane (2025), who tested whether headtracking during 360 degree VR videos could distinguish people based on depression severity. Their main findings with 50 participants were:

- People with moderate to severe depression showed significantly slower scanning speeds (M = 3.88 degrees/s, SD = 0.88) compared to mild (M = 4.94) and no depression (M = 5.29) groups
- The effect sizes were large (Cohen's d = 1.22 to 1.64)
- These effects held after controlling for anxiety using ANCOVA ($p < .001$, partial eta squared = .295)
- Self-reported valence and arousal showed no group differences. Only the objective headtracking data could differentiate the groups

1.3 Our Study

We follow the same paradigm with N = 40 college students (36 male, 4 female, ages 19 to 27, M = 22.78, SD = 1.80) using a Meta Quest 3 VR headset. Participants viewed 5 different 360 degree videos spanning diverse emotional conditions, completed standardized questionnaires (PHQ-9, GAD-7, STAI-T, PANAS), and rated each video on valence and arousal.

1.4 Hypotheses

H1: People with higher PHQ-9 depression scores will show lower headtracking scanning speeds during 360 degree video exploration.

H2: This effect will be strongest for yaw (horizontal scanning) speed, matching the reference study.

H3: The depression effect on scanning speed will be consistent across video types, reflecting general psychomotor retardation rather than stimulus-specific emotional blunting.

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2. Methods

2.1 Participants and Design

Component	Detail
Participants	N = 40 college students (36 male, 4 female), ages 19 to 27 (M = 22.78, SD = 1.80)
VR Headset	Meta Quest 3
Videos	5 different 360 degree videos spanning different emotional conditions
Surveys	PHQ-9, GAD-7, STAI-T, PANAS (pre and post VR), SAM valence/arousal per video, Presence, VRISE
Headtracking	Rotation data (Pitch/X, Yaw/Y, Roll/Z) plus speed, recorded per video per participant
Design	Mixed design with between-subjects factor (PHQ-9 group, 3 levels) and within-subjects factor (video type, 5 levels)

2.2 Depression Group Classification

We classified participants using established PHQ-9 clinical cutoffs, following the reference paper:

PHQ-9 Group	Score Range	N	Percentage
Minimal	0 to 4	20	50%
Mild	5 to 9	12	30%
Moderate-Severe	10 or above	8	20%

Limitation: Our moderate-severe group (n = 8) is much smaller than in the reference paper (n = 20, 40% of their sample), which directly impacts statistical power.

2.3 Videos

Video	Content	Duration	Emotional Condition
V1	Abandoned Buildings	60s	Eerie/Unsettling
V2	Evening at the Beach	60s	Pleasant/Calm
V3	Campus	60s	Familiar/Neutral
V4	The Nun Horror	about 3 min	Fearful/Intense
V5	Tahiti Surf	about 3 min	Pleasant/Exciting

2.4 Measures

- **PHQ-9**: 9 item depression screener, scores range from 0 to 27 (Kroenke et al., 2001)
- **GAD-7**: 7 item anxiety screener, scores range from 0 to 21 (Spitzer et al., 2006)
- **STAI-T**: 20 item trait anxiety measure, scores range from 20 to 80
- **PANAS**: 20 item positive/negative affect measure, given before and after VR (Watson et al., 1988)
- **Headtracking**: Rotational speed (total, X/pitch, Y/yaw, Z/roll) and standard deviation of yaw (SDY) per video

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3. Preliminary Analysis Summary (Report 1)

Report 1 was the midterm submission where we presented descriptive statistics, group distributions, and preliminary inferential tests. Below is a summary of those findings.

3.1 Sample Characteristics

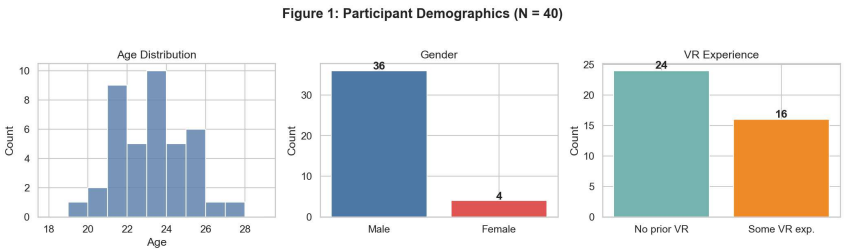

Figure 1: Participant Demographics

Our sample was mostly male (90%) and college-aged, with limited prior VR experience. Depression and anxiety scores showed the expected right-skewed distribution:

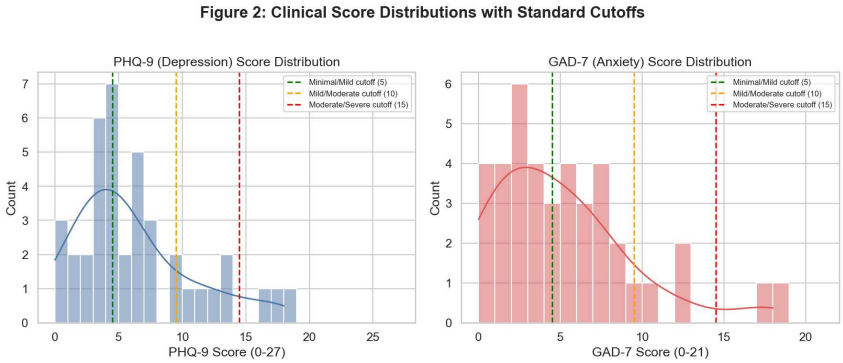

Figure 2: Clinical Score Distributions

The strong correlation between PHQ-9 and GAD-7 (Spearman rho = .710, p < .001) motivated a distress grouping approach, consistent with the reference paper.

3.2 VR Experience Effects

- Positive affect decreased significantly after VR exposure (Wilcoxon signed-rank test, W = 236, p = .031). *Wilcoxon was used because the paired pre-post differences were not normally distributed.*
- Negative affect did not change significantly (W = 235, p = .414)
- The videos successfully covered different emotional conditions (V4 Horror had the lowest valence and highest arousal, V5 Surf had the highest valence)

3.3 Preliminary Headtracking Results

Welch's t-tests comparing the Minimal and Moderate-Severe depression groups found no significant differences in scanning speed for any video (all p values above .27). *Welch's t-test was chosen over the standard t-test because it does not assume equal variances between the two groups.* The largest effects, though still not significant, were in the expected direction for Tahiti Surf (d = 0.44) and Campus (d = 0.37). *Cohen's d measures the standardized difference between two group means, where values of 0.2, 0.5, and 0.8 are considered small, medium, and large effects.*

One-way ANOVA was used to compare scanning speed across all three PHQ-9 groups simultaneously (rather than doing multiple t-tests). Results: Total Speed F(2,37) = 0.644, p = .531 and Yaw Speed F(2,37) = 0.618, p = .545. Both non-significant. *F is the ratio of between-group variance to within-group variance. A larger F means bigger group differences relative to individual differences.*

Spearman rank correlations between PHQ-9 and scanning speed were weak and non-significant (rho = .026, p = .875 for total speed). *Spearman was used instead of Pearson because PHQ-9 scores are not normally distributed. Spearman's rho measures the strength of the monotonic relationship between two variables, ranging from -1 to +1.*

3.4 Report 1 Conclusions

The preliminary analysis found no statistically significant headtracking differences between depression groups. This was likely due to the smaller moderate-severe sample (n = 8 vs. n = 20 in the reference paper). Report 2 extends these findings with formal normality checks, ANCOVA, regression, mixed ANOVA, and power analysis.

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4. Statistical Analysis (Report 2)

4.1 Analytical Workflow

Our analysis followed a step-by-step pipeline:

1. **Normality assessment** using Shapiro-Wilk tests and Q-Q plots to decide between parametric and non-parametric tests
2. **ANCOVA** to test the depression group effect on scanning speed while controlling for anxiety (GAD-7), following the reference paper's main analysis

3. **Multiple regression** using PHQ-9 as a continuous predictor to avoid losing information from grouping
4. **Two-way mixed ANOVA** to test whether video type interacts with depression group
5. **SDY analysis** using standard deviation of yaw as a complementary metric
6. **Post-hoc power analysis** to formally assess our study's detection ability

4.2 Normality Assessment

We used the Shapiro-Wilk test to check whether each variable follows a normal distribution. *This test is recommended for small samples (N < 50) and produces a W statistic close to 1 for normal data. If p < .05, the variable is considered non-normal.*

Table 1: Shapiro-Wilk Normality Test Results (Overall Sample)

Variable	N	W	p	Normal?	Skewness
PHQ-9	40	0.898	.002	No	1.022
GAD-7	40	0.883	.001	No	1.273
STAI-T	40	0.944	.045	No	0.371
Positive Affect (Pre)	40	0.977	.585	Yes	0.135
Negative Affect (Pre)	40	0.729	<.001	No	2.226
Avg Scan Speed (Total)	40	0.978	.602	**Yes**	0.026
Avg Scan Speed (Yaw)	40	0.976	.537	**Yes**	0.024

The main dependent variables, scanning speed (total and yaw), passed the normality test (W above .975, p above .50), supporting the use of parametric tests. The clinical variables (PHQ-9, GAD-7) were positively skewed, as expected in a non-clinical population.

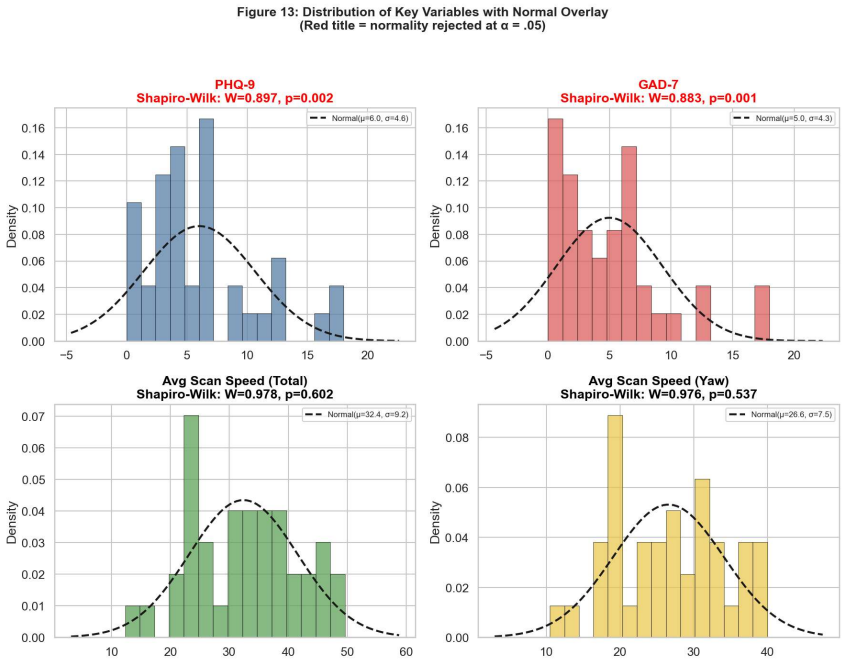

Figure 3: Distributions with Normal Overlay

All within-group scanning speed distributions were also normal (all Shapiro-Wilk p above .20), confirming that ANOVA and t-test assumptions are met.

Corrections to Report 1 Test Choices

The normality results showed that some test choices in Report 1 were not ideal:

Report 1 Test	Correction	Reasoning
Kruskal-Wallis (speed across groups)	**One-way ANOVA**	Speed is normal within all groups, so ANOVA gives more power
Pearson correlations (PHQ-9/GAD-7 vs speed)	**Spearman rank**	PHQ-9 and GAD-7 are not normally distributed (right-skewed)
Paired t-test (PANAS pre vs post)	**Wilcoxon signed-rank**	The paired differences do not follow a normal distribution

Note: These corrections did not change any of the conclusions from Report 1.

4.3 ANCOVA: Depression Effect Controlling for Anxiety

We used ANCOVA (Analysis of Covariance) because depression and anxiety are strongly related (r = .579 in our sample), and we need to separate the effect of depression from anxiety. ANCOVA lets us test whether depression groups differ in scanning speed while statistically removing the influence of GAD-7 anxiety scores. *Partial eta squared measures how much variance in the outcome is explained by the group factor after accounting for the covariate. Values of .01, .06, and .14 are considered small, medium, and large.*

Assumptions checked: Normality within groups (passed), Levene's test for equal variances (p = .119, passed), homogeneity of regression slopes (F = 0.043, p = .958, passed).

Table 2: ANCOVA Results. Scanning Speed by PHQ Group, Controlling for GAD-7

DV	Source	df	F	p	Partial eta squared
Total Speed	PHQ Group	2	0.855	.434	.045
	GAD-7 (covariate)	1	0.527	.473	
Yaw Speed	PHQ Group	2	0.883	.422	.047
	GAD-7 (covariate)	1	0.606	.441	

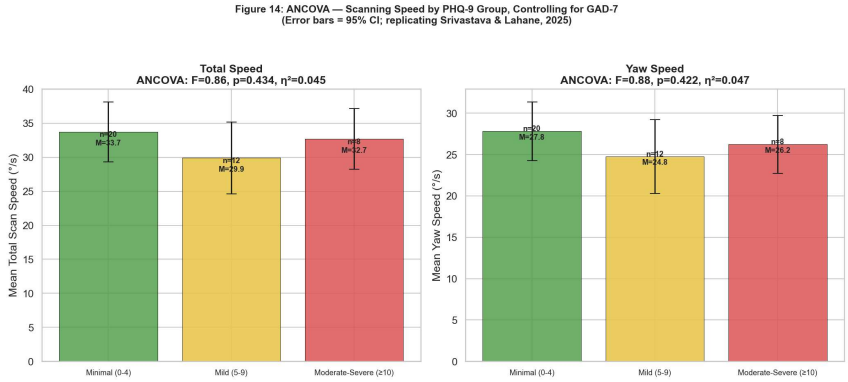

Figure 4: ANCOVA-Adjusted Group Means

There was no significant effect of PHQ-9 group on scanning speed after controlling for anxiety. This was true for both total speed (p = .434, partial eta squared = .045) and yaw speed (p = .422, partial eta squared = .047). Depression group membership explains less than 5% of the variance in scanning speed, which is a big contrast with the reference paper's partial eta squared of .295.

4.4 Linear Regression: PHQ-9 as Continuous Predictor

We used linear regression because splitting participants into groups (like we did for ANCOVA) can lose statistical information. Regression treats PHQ-9 as a continuous number, preserving more information and giving us more power to detect a relationship. *R squared tells us what percentage of the variance in scanning speed is explained by the predictors. Beta is the slope, showing how much scanning speed changes per unit increase in PHQ-9. VIF (Variance Inflation Factor) checks if predictors are too correlated with each other; values below 2.5 are acceptable.*

Table 3: Regression Models

Model	DV	Predictors	R squared	PHQ-9 Beta	PHQ-9 p
1	Total Speed	PHQ-9	.004	0.168	.712
2	Total Speed	PHQ-9 + GAD-7	.005	0.282	.622
3	Total Speed	PHQ-9 + GAD-7 + STAI-T	.006	0.250	.703
4	Yaw Speed	PHQ-9 + GAD-7	.008	0.227	.600

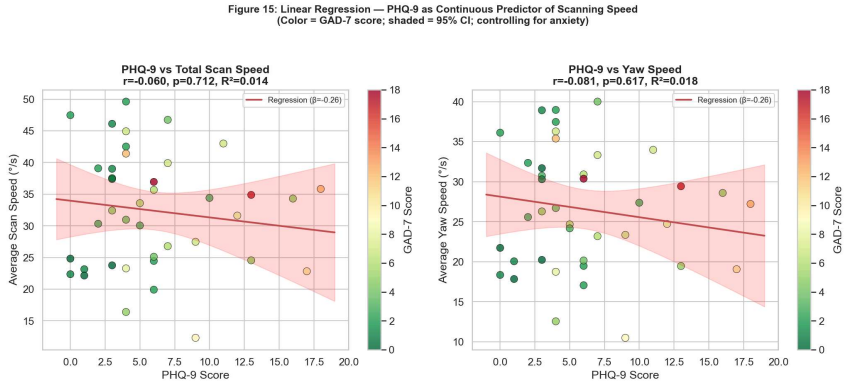

Figure 5: PHQ-9 vs Scanning Speed (Regression Scatter)

PHQ-9 was not a significant predictor in any model (all p above .60). The R squared values were close to zero (about 0.5%), meaning depression severity explains almost none of the variance in scanning speed. We also checked all regression assumptions: residual normality (W = 0.978, p = .602), homoscedasticity (Breusch-Pagan LM = 0.477, p = .788), multicollinearity (all VIF below 2.5), and no influential outliers.

4.5 Two-Way Mixed ANOVA: Video x Depression Interaction

We used a two-way mixed ANOVA to test two things at once: whether depression groups differ in scanning speed (between-subjects effect), and whether this difference depends on which video is being watched (interaction effect). It is called "mixed" because it combines a between-subjects factor (PHQ group, which differs across people) with a within-subjects factor (video type, which every participant experiences). *Sphericity is an assumption that the variances of the differences between all pairs of within-subject conditions are equal. Mauchly's test checks this.*

Sphericity: Mauchly's W = 0.740, p above .05. The assumption was met, so no correction was needed.

Table 4: Mixed ANOVA Results

Source	df1	df2	F	p	Partial eta squared
PHQ Group (between)	2	37	0.644	.531	.034
Video (within)	**4**	**148**	**34.993**	**<.001**	**.486**
Interaction	8	148	1.233	.284	.062

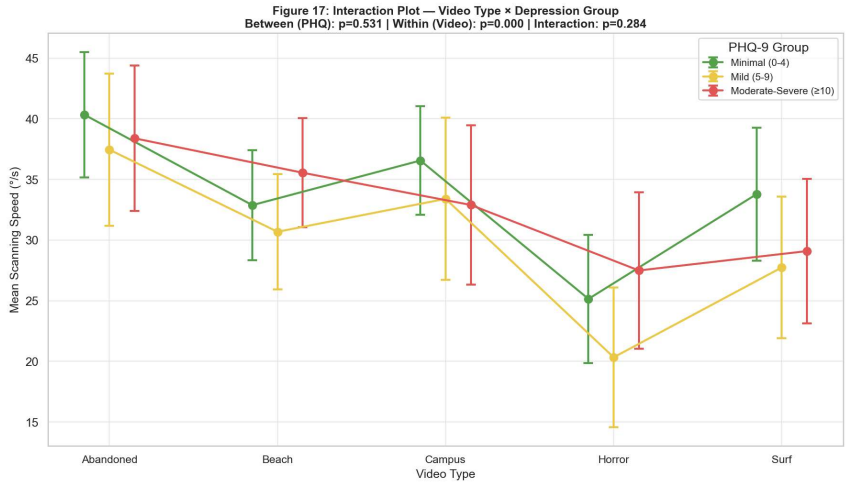

Figure 6: Video x Depression Group Interaction

Video type had a strong effect on scanning speed ($p < .001$, partial eta squared = .486), which is expected since different videos naturally cause different amounts of head movement. However, there was no significant main effect of depression ($p = .531$) and no significant interaction ($p = .284$). Depression did not affect scanning differently across video types. This matches the reference paper's pattern (uniform effect), though in our case the overall depression effect itself was not significant.

4.6 Standard Deviation of Yaw (SDY) Analysis

SDY measures the range of horizontal exploration (how widely people look around), which complements scanning speed (how fast they look around). We used the Kruskal-Wallis test here because SDY did not pass the normality check in several video conditions. *Kruskal-Wallis is the non-parametric alternative to one-way ANOVA. It compares group medians using ranks instead of means, and does not require normality.*

Table SDY: SDY by Depression Group

PHQ-9 Group	N	SDY Mean	SDY SD
Minimal	20	76.94	11.05
Mild	12	76.70	16.10
Moderate-Severe	8	82.06	10.90

Kruskal-Wallis test: $H(2) = 1.629$, $p = .443$. No significant group difference. Correlations with PHQ-9 were also not significant (Spearman rho = .121, $p = .459$). One interesting finding: the Horror video showed a nearly significant SDY difference (Minimal M = 28.82 vs. Moderate-Severe M = 49.06, $t = 2.27$, $p = .053$, $d = 1.086$), but in the opposite direction. The more depressed group showed more yaw variability, which might reflect anxious scanning (heightened vigilance) rather than psychomotor retardation.

4.7 Power Analysis

Since we found no significant results, it is important to check whether our study had enough statistical power to detect the effects reported in the reference paper. *Statistical power is the probability of correctly detecting a real effect if one exists. A power of .80 (80%) is the standard threshold. Power depends on sample size, effect size, and significance level (alpha = .05).*

Table 5: Post-Hoc Power with Our Sample Sizes

Comparison	d (reference)	n1	n2	Power
Minimal vs Moderate-Severe	1.64	20	8	**.965**
Mild vs Moderate-Severe	1.22	12	8	.715
Minimal vs Mild	0.37	20	12	.165

Our observed effect was $d = 0.125$, with post-hoc power of only 0.060, which is essentially at chance level.

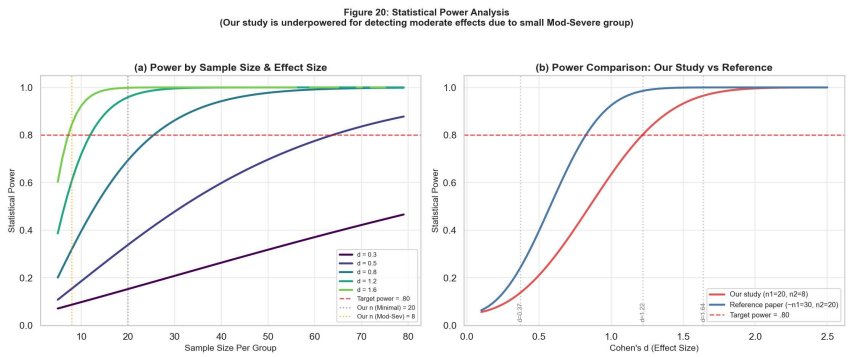

Figure 7: Power Analysis Curves

This is an important finding. We had 96.5% power to detect the reference paper's large effect ($d = 1.64$) for the Minimal vs. Moderate-Severe comparison. This means if the true effect were as large as they reported, we almost certainly would have found it. Since we did not, this suggests:

1. The reference paper's effect sizes may have been overestimated, possibly due to sample-specific factors or publication bias

2. There may be population differences between our sample (Indian technical university students) and the reference paper's sample
3. With only 20% moderate-severe participants (vs. 40% in the reference), our moderate-severe group may have had milder cases on average

4.8 Extended Correlations

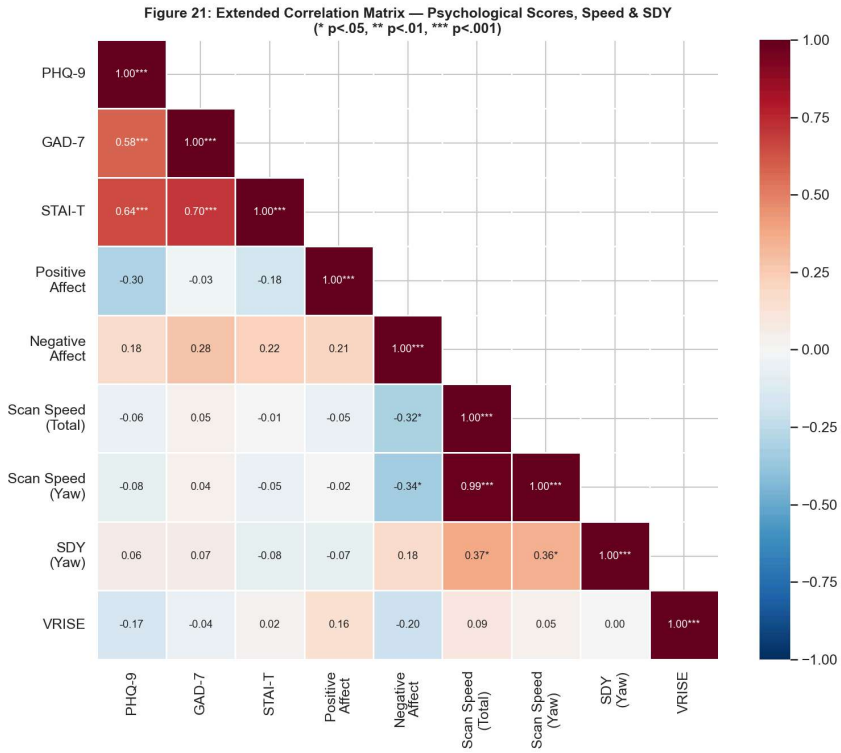

Figure 8: Extended Correlation Matrix

Table 6: Notable Correlations

Variable Pair	r	p
PHQ-9 and GAD-7	.579	<.001
PHQ-9 and STAI-T	.642	<.001
Negative Affect and Total Speed	**-.320**	**.044**
Negative Affect and Yaw Speed	**-.341**	**.031**
Total Speed and SDY	.374	.017

An interesting finding: while PHQ-9 did not correlate with scanning speed, pre-VR negative affect (from PANAS) did show a significant negative correlation with both total speed (r = -.320, p = .044) and yaw speed (r = -.341, p = .031). This suggests that how someone feels right now (current mood state) may be a better predictor of VR scanning behavior than their depression score over the past two weeks (trait-level depression), especially in non-clinical samples like ours.

5. Conclusions

5.1 Summary of Findings

1. **Normality**: Scanning speed variables are normally distributed, which supports the use of parametric methods. Clinical variables (PHQ-9, GAD-7) show the expected positive skew.
2. **ANCOVA**: No significant effect of depression on scanning speed after controlling for anxiety (p = .434, partial eta squared = .045).
3. **Regression**: PHQ-9 explains less than 1% of variance in scanning speed (R squared is about .005).
4. **Mixed ANOVA**: Video type strongly affects scanning speed (p < .001, partial eta squared = .486), but neither the depression main effect (p = .531) nor the interaction (p = .284) is significant.
5. **SDY**: No significant group differences in yaw variability (p = .443).

6. **Power**: We had 96.5% power to detect the reference paper's effect size of d = 1.64, yet our observed effect was only d = 0.13. This suggests the original effects may be inflated or specific to that population.
7. **Negative affect**: Pre-VR negative affect significantly correlates with reduced scanning speed (r = -.32, p = .044), suggesting state-level mood may be a better predictor of scanning behavior than trait depression.

5.2 Assessment

Our study did not find the significant headtracking-depression relationship reported by Srivastava and Lahane (2025). Even though we followed the same paradigm, used similar stimuli, and had a comparable sample size, none of our analyses (ANCOVA, regression, mixed ANOVA, SDY) produced significant results. The power analysis confirms this is not simply a matter of low power for detecting large effects.

5.3 Implications and Future Directions

1. **Larger, clinically diverse samples**: Recruiting from clinical populations would directly test whether the reference paper's effects hold in people with more severe depression.
2. **State vs. trait measures**: Our finding that negative affect correlates with scanning speed while PHQ-9 does not suggests VR scanning may be more sensitive to how someone feels right now than to chronic depression.
3. **Temporal dynamics**: Looking at how scanning speed changes over the course of each video (trajectory analysis) might reveal depression-related patterns that simple averages miss.
4. **Anxiety-specific effects**: The nearly significant SDY finding during the Horror video suggests anxiety may play its own role in VR scanning behavior, especially with threatening content.

References

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